

C618

Earth Science: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A geologist with a state survey maps a canyon wall: layer A at the bottom, then B, then C on top. A basalt dike D cuts through A and B but ends at the base of C. The canyon is popular with hikers. What is the order of events, oldest first?

- A. C, B, A, D
- B. A, B, D, C
- C. D, A, B, C
- D. A, B, C, D

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A fault cuts layers 1 through 4 (1 is lowest) and is covered by unbroken layer 5. What is the fault's relative age?

- A. Older than all five layers
- B. Younger than all five layers
- C. It cannot be determined without radiometric dating
- D. Younger than layer 4 and older than layer 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A museum with a famous mammoth skeleton tests a bone sample and finds 25% of its original carbon-14 remains. Carbon-14 has a half-life of 5,730 years. The museum hosts 200,000 visitors a year. About how old is the bone?

- A. About 4,300 years
- B. About 17,190 years
- C. About 11,460 years
- D. About 5,730 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An isotope has a half-life of 1.3 billion years. A rock is 3.9 billion years old. What fraction of the original parent isotope remains?

- A. $\frac{1}{8}$
- B. $\frac{1}{6}$
- C. $\frac{1}{3}$
- D. $\frac{7}{8}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

After a large earthquake recorded by 900 stations worldwide, seismologists note that S waves are not detected on the far side of Earth. The earthquake happened at 3 a.m. local time. What does this show?

- A. The outer core is liquid.
- B. The inner core is liquid.
- C. The mantle is liquid magma.
- D. S waves are too slow to reach that far.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which type of seismic wave can pass through Earth's outer core?

- A. S waves
- B. None of them
- C. P waves
- D. Surface waves

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Moist air from the ocean is pushed up the windward slope of a mountain range with a ski resort at the top. The resort employs 300 seasonal workers. Why do clouds form as the air rises?

- A. Higher air is closer to the Sun, so more water evaporates.
- B. The mountain adds moisture to the air.
- C. The air expands, cools to its dew point, and water vapor condenses.
- D. The air is compressed and warms, which releases water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An ocean research ship with a crew of 30 samples the North Atlantic near Greenland, where surface water sinks to form deep water. The ship was built in Norway. What mainly causes this water to sink?

- A. It is warm and fresh.
- B. Strong winds push it down.
- C. The Moon's tides pull it down.
- D. It is cold and salty, so it is dense.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A teacher who leads a climate club with 18 members asks why rising CO₂ warms Earth's surface. The club meets in a portable classroom. Which answer is correct?

- A. CO₂ absorbs infrared energy radiated by the surface and re-emits some back down.
- B. All atmospheric gases trap heat equally.
- C. CO₂ creates a hole in the ozone layer that lets heat in.
- D. CO₂ blocks incoming sunlight from reaching the ground.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An astronomy club at a college with a small observatory studies a main-sequence star with 10 times the Sun's mass. The club's telescope was donated by an alumnus. How does the star's lifetime compare with the Sun's?

- A. About 10 times longer, since it has 10 times the fuel
- B. Much shorter—hundreds of times shorter
- C. The same; all main-sequence stars last about 10 billion years
- D. Exactly 10 times shorter

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A star ends its life as a black hole. What was most likely true of the star?

- A. Any star becomes a black hole if it is old enough.
- B. It had about the Sun's mass.
- C. It had less mass than the Sun.
- D. It was many times more massive than the Sun.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A planetarium director who gives 12 shows a week describes an asteroid orbiting the Sun at an average distance of 4 AU. The planetarium's dome is 20 meters wide. How long does the asteroid take to orbit the Sun?

- A.** 64 years
- B.** 8 years
- C.** 4 years
- D.** 16 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A geologist ~~with a state survey~~ maps a canyon wall: layer A at the bottom, then B, then C on top. A basalt dike D cuts through A and B but ends at the base of C. ~~The canyon is popular with hikers.~~ What is the order of events, oldest first?

Target word: sequence of events **Decides:** Dike cuts A and B, but C covers it.

A	Inverted operation	Reverses superposition: the bottom layer is oldest.
B	Correct	Superposition orders the layers; D cuts A and B, then C was deposited on top.
C	Confusion set	Treats the intrusion as older than the rocks it cuts.
D	Dropped element	Ignores that C lies unbroken over the dike.

2. Answer D Concept 1 • transfer

A fault cuts layers 1 through 4 (1 is lowest) and is covered by unbroken layer 5. What is the fault's relative age?

Target word: bracket the fault's age **Decides:** Cross-cutting plus superposition.

A	Inverted operation	A cutting feature is younger than what it cuts.
B	Dropped element	Ignores that layer 5 is unbroken.
C	Absolute language	Relative age can be found from the layers alone.
D	Correct	It cuts 4 and is covered by 5.

3. Answer C Concept 2

A museum ~~with a famous mammoth skeleton~~ tests a bone sample and finds 25% of its original carbon-14 remains. Carbon-14 has a half-life of 5,730 years. ~~The museum hosts 200,000 visitors a year.~~ About how old is the bone?

Target word: age from fraction remaining **Decides:** 25% = two half-lives.

A	Inverted operation	Treats decay as linear: 75% of one half-life.
B	Confusion set	Counts three half-lives.
C	Correct	100% → 50% → 25% is two half-lives: $2 \times 5,730$.
D	Dropped element	One half-life leaves 50%, not 25%.

4. Answer A Concept 2 • transfer

An isotope has a half-life of 1.3 billion years. A rock is 3.9 billion years old. What fraction of the original parent isotope remains?

Target word: fraction remaining after three half-lives **Decides:** $3.9 \div 1.3 = 3$ half-lives.

A	Correct	$(1/2)^3 = 1/8$.
B	Inverted operation	Multiplies 1/2 by 1/3 instead of raising to the third power.
C	Confusion set	Treats decay as linear.
D	Adjacent category	That is the fraction that has decayed into daughter product.

5. Answer A Concept 3

After a large earthquake ~~recorded by 900 stations worldwide~~, seismologists note that S waves are not detected on the far side of Earth. ~~The earthquake happened at 3 a.m. local time.~~ What does this show?

Target word: what the S-wave shadow shows **Decides:** S waves cannot cross liquid.

A	Correct	S waves stop where they reach the liquid outer core.
B	Confusion set	The inner core is solid; the liquid layer is the outer core.
C	Adjacent category	The mantle is solid rock that can flow slowly; S waves cross it.
D	Dropped element	Speed is not the reason; the medium blocks them.

6. Answer C Concept 3 • transfer

Which type of seismic wave can pass through Earth's outer core?

Target word: reverse: which wave crosses liquid **Decides:** P waves are compressional.

A	Inverted operation	S waves cannot travel through liquid.
B	Absolute language	P waves do pass through, though they bend.
C	Correct	Compressional waves travel through solids and liquids.
D	Adjacent category	They travel along the surface, not through the core.

7. Answer C Concept 4

Moist air from the ocean is pushed up the windward slope of a mountain range ~~with a ski resort at the top. The resort employs 300 seasonal workers.~~ Why do clouds form as the air rises?

Target word: cause of cloud formation **Decides:** Expansion cooling to dew point.

A	Confusion set	Altitude brings cooling, not more solar heating.
B	Adjacent category	The moisture came from the ocean.
C	Correct	Lower pressure aloft causes adiabatic cooling.
D	Inverted operation	Rising air expands and cools.

8. Answer D Concept 5

An ocean research ship ~~with a crew of 30~~ samples the North Atlantic near Greenland, where surface water sinks to form deep water. ~~The ship was built in Norway.~~ What mainly causes this water to sink?

Target word: driver of deep water formation **Decides:** Density from cold and salinity.

A	Inverted operation	Warm, fresh water is less dense and stays near the surface.
B	Adjacent category	Wind drives surface currents, not deep sinking.
C	Confusion set	Tides move water up and down daily; they do not form deep water.
D	Correct	Thermohaline circulation is density-driven.

9. Answer A Concept 6

A teacher who leads a climate club ~~with 18 members~~ asks why rising CO₂ warms Earth's surface. ~~The club meets in a portable classroom.~~ Which answer is correct?

Target word: mechanism of greenhouse warming **Decides:** CO₂ absorbs outgoing infrared.

A	Correct	Greenhouse gases slow heat loss to space.
B	Absolute language	Nitrogen and oxygen absorb little infrared.
C	Confusion set	Ozone depletion is a separate problem involving UV.
D	Inverted operation	That would cool the surface, not warm it.

10. Answer B Concept 7

An astronomy club ~~at a college with a small observatory~~ studies a main-sequence star with 10 times the Sun's mass. ~~The club's telescope was donated by an alumnus.~~ How does the star's lifetime compare with the Sun's?

Target word: lifetime vs. mass **Decides:** Massive stars burn fuel much faster.

A	Dropped element	Drops the much faster burn rate.
B	Correct	Fuel use rises far faster than fuel supply as mass increases.
C	Absolute language	Lifetime depends strongly on mass.
D	Wrong stem number	Uses the mass ratio directly; the drop is far larger.

11. Answer D Concept 7 · transfer

A star ends its life as a black hole. What was most likely true of the star?

Target word: reverse: mass from the endpoint **Decides:** Only very massive stars collapse to black holes.

A	Absolute language	The endpoint depends on mass, not age.
B	Confusion set	Sun-like stars end as white dwarfs.
C	Inverted operation	Low-mass stars end as white dwarfs.
D	Correct	Only the most massive stars leave black holes.

12. Answer B Concept 8

A planetarium director ~~who gives 12 shows a week~~ describes an asteroid orbiting the Sun at an average distance of 4 AU. ~~The planetarium's dome is 20 meters wide.~~ How long does the asteroid take to orbit the Sun?

Target word: orbital period **Decides:** $P^2 = a^3 = 64$.

A	Inverted operation	Stops at a^3 without taking the square root.
B	Correct	$\sqrt{64} = 8$.
C	Dropped element	Sets P equal to a.
D	Confusion set	Uses $P = a^2$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Relative dating principles	In undisturbed layers the oldest is at the bottom (superposition); a feature that cuts across rocks is younger than the rocks it cuts (cross-cutting).	Treating an intrusion or fault as older than the layers it cuts.	A dike or fault drawn through some layers but not others.
2	Radiometric dating and half-life	After each half-life, half the remaining parent isotope decays; after n half-lives, $(1/2)^n$ of the parent remains.	Treating decay as linear.	A percent or fraction of parent isotope remaining.
3	Seismic evidence for Earth's interior	S waves cannot travel through liquid, so the S-wave shadow zone shows the outer core is liquid; P waves pass through solids and liquids.	Thinking the mantle is liquid magma or that S waves pass through the core.	A shadow zone or "which wave passes through".
4	Adiabatic cooling and clouds	Rising air expands under lower pressure and cools; when it cools to its dew point, water vapor condenses into clouds.	Believing rising air warms or that altitude brings air closer to the Sun's heat.	Air forced up a mountain or along a front.
5	Thermohaline circulation	Deep ocean currents are driven by density differences: cold, salty water sinks, while surface currents are mainly wind-driven.	Assigning deep currents to wind or tides.	Deep water forming in polar seas.
6	The greenhouse effect	Greenhouse gases such as CO_2 and water vapor let sunlight in but absorb and re-emit outgoing infrared, warming the surface; the ozone hole is a separate issue.	Confusing the greenhouse effect with ozone depletion.	CO_2 , infrared, or "why does the surface warm?"
7	Stellar mass controls a star's life	More massive stars burn fuel much faster and live far shorter lives; only very massive stars end as neutron stars or black holes.	Thinking more fuel means a longer life.	A star several times the Sun's mass.
8	Kepler's third law	For objects orbiting the Sun, $P^2 = a^3$ with P in years and a in astronomical units.	Using $P = a$ or $P = a^2$ instead of the square root of a^3 .	An orbital distance in AU and "how long is its year?"

Course scope (public description): This course covers the advanced content knowledge that a secondary earth science teacher is expected to know and understand. Topics include basic scientific principles of earth and space sciences, tectonics and internal earth processes, earth materials and surface processes, history of the Earth and its life-forms, Earth's atmosphere and hydrosphere, and astronomy.